Name:_____ Class: _____ Date: _____

How to Balance Chemical Equations

The Law of Conservation of Mass states that matter is never created or destroyed. When writing chemical equations, the Law of Conservation of Mass has to be satisfied. The number and kind of atoms contained in the reactants must equal the number and kind of atoms in the products. If the atoms are not equal, we have defied the Law of Conservation of Mass. Therefore, we must balance the equation.

Balancing chemical equations is easy if you follow some rules.

- You can only add a coefficient in front of a chemical formula
- You cannot change any subscripts, nor add any subscripts
- You may not place a coefficient in the 'middle' of a chemical formula
- You must end up with the *same number and kinds of atoms* on both sides of the equation when you are finished. Count atoms on both sides upon completion to check yourself.

Practice with this example.

$CO + H_2 \longrightarrow CH_3OH$

Reactants are on the left side of the arrow - products are on the right.

Count the number of atoms on the left side of the arrow, and compare it with the number of atoms on the right side.

If you draw a vertical line under the arrow, you can easily see the two sides of your equation. List the atoms as shown.

$CO + H_2 \longrightarrow CH_3OH$

C - 1	C – 1
O – 1	O – 1
H – 2	H – 4

If you list the atoms in the same order on both sides, you can easily see which atoms are equal and which are not. Here, you see that we do not have the same number of hydrogen atoms on both sides. We must follow the rules above to balance this equation. Add a coefficient of 2 in front of the H_2, but do not change any subscripts.

$CO + 2H_2 \longrightarrow CH_3OH$

For every molecule of CO, we must use 2 molecules of H_2 to produce 1 molecule of CH_3OH.

The $2H_2$ means there are 2 molecules of H_2 – therefore there are 4 atoms of hydrogen. Recheck the number of atoms for accuracy. The number of atoms on both sides of the equation is now equal. We have satisfied the Law of Conservation of Mass.

Name:_____ Class: _____ Date: _____

Here's another example:

$Al + O_2 \longrightarrow Al_2O_3$

Al – 1	Al – 2
O – 2	O – 3

Count the number of atoms and list them under the equation. Compare each side.

You might try adding a coefficient of 2 in front of the Al to balance the aluminum atoms.

$2Al + O_2 \longrightarrow Al_2O_3$

Al – 2	Al – 2
O – 2	O - 3

It's OK if the first coefficient you try does not balance the equation.

But: How should we fix the number of oxygen atoms? By looking at the subscripts for oxygen, you see that we cannot get them to match without adding coefficients. If we add a coefficient of 2 in front of the O_2 on the left, we will then have 4 oxygen atoms. We cannot achieve 4 oxygen atoms on the right using a coefficient. Let's try the lowest common multiple of 2 and 3. That number is 6. We will attempt to have 6 oxygen atoms on both sides of the equation.

To do this, add a coefficient of 3 in front of the oxygen on the left, and a coefficient of 2 in front of the Al_2O_3 on the right. Remember, you cannot add a coefficient in between a molecular formula.

$2Al + 3O_2 \longrightarrow 2Al_2O_3$

Al – 2	Al – 4
O – 6	O – 6

Now we have balanced the number of oxygen atoms in our equation, but have caused the aluminum atoms to be unbalanced.

- The coefficient is multiplied by each subscript to get the total number of atoms.

Replace the '2' in front of the Al with a '4' to balance the number of aluminum atoms.

$4Al + 3O_2 \longrightarrow 2Al_2O_3$

Now we have a balanced equation.

Difficulties encountered when balancing equations like the one above may be solved if you are patient, follow the balancing rules, count numbers of atoms correctly, and give the 'trial and error' method a try.

Name:_____ Class: _____ Date: _____

Balancing Chemical Equations

Practice balancing these equations. Start by drawing the dividing line and counting atoms. Be sure to follow the Law of Conservation of Mass. Check your answers.

1. $Zn + HCl \longrightarrow ZnCl_2 + H_2$

2. $C_{12}H_{22}O_{11} \longrightarrow C + H_2O$

3. $Fe + O_2 \longrightarrow Fe_3O_4$

4. $SnO_2 + C \longrightarrow CO + Sn$

Name:_____ Class: _____ Date: _____

5. $Na_2CO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$

6. $CaC_2 + H_2O \longrightarrow C_2H_2 + Ca(OH)_2$

7. $KClO_3 \longrightarrow KClO_4 + KCl$

Name:_____ Class: _____ Date: _____

Balancing Chemical Equations

Answer Key

Practice balancing these equations. Start by drawing the dividing line and counting atoms. Be sure to follow the Law of Conservation of Mass. Check your answers.

1. $Zn + HCl \longrightarrow ZnCl_2 + H_2$
   $Zn + 2HCl \longrightarrow ZnCl_2 + H_2$

2. $C_{12}H_{22}O_{11} \longrightarrow C + H_2O$
   $C_{12}H_{22}O_{11} \longrightarrow 12C + 11H_2O$

3. $Fe + O_2 \longrightarrow Fe_3O_4$
   $3Fe + 2O_2 \longrightarrow Fe_3O_4$

4. $SnO_2 + C \longrightarrow CO + Sn$
   $SnO_2 + 2C \longrightarrow 2CO + Sn$

5. $Na_2CO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$
   $Na_2CO_3 + 2HCl \longrightarrow 2\,NaCl + H_2O + CO_2$

6. $CaC_2 + H_2O \longrightarrow C_2H_2 + Ca(OH)_2$
   $CaC_2 + 2H_2O \longrightarrow C_2H_2 + Ca(OH)_2$

7. $KClO_3 \longrightarrow KClO_4 + KCl$
   $4KClO_3 \longrightarrow 3KClO_4 + KCl$

Name:_____ Class: _____ Date: _____

Teacher Notes

This worksheet provides the steps to balancing chemical equations and provides practice for the students. The first portion of this worksheet is a set of instructions for how to balance an equation. This information is also covered on the power point slide show accompanying this download and may not be needed, unless you would like to give these instructions to a student who was absent, or a student who struggles with note-taking.

Have students practice their balancing skills by giving each student a copy of this handout and having them complete it. It could be assigned as a homework, individual class work, or group work.